

Stem Cells and Meristems: Potential Compared

I can compare embryonic stem cells, adult tissue stem cells and plant meristem cells using evidence about self-renewal, differentiation range and function.

SELFRENEW

make more stem cells

MERISTEM

plant growth region

POTENTIAL

range of cell types

[Open the original interactive lesson](#)

What success looks like

- Define a stem cell as an unspecialised cell that can divide and differentiate.
- Compare the wider differentiation potential of embryonic stem cells with the more limited range of many adult tissue stem cells.
- Explain how meristem cells support continued plant growth and can form specialised plant tissues.

Retrieve and reconnect

- Define differentiation.
- State one role of cell division in growth or repair.
- Name one process occurring in a plant meristem.
- Lead-in: predict why an adult tissue keeps a small stem-cell population.

Which claim is accurate enough to test against the comparison evidence?

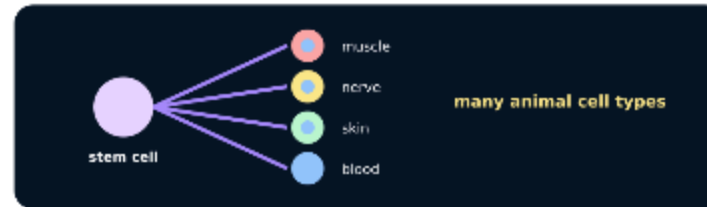
Think first. Point, say, sign or write your choice. Give one reason.

- A. Every stem cell can become absolutely any cell in any organism.
- B. Different stem-cell sources can have different differentiation ranges and biological roles.
- C. A stem cell is already fully specialised and cannot divide.

Look closely · what is the evidence?



A stem cell dividing into many cell types



Embryonic range



Adult and meristem

Words we will use

- stem cell — unspecialised source cell
- self-renewal — maintain the source
- meristem — plant growth tissue

Watch the model

- Use a branching model: one route maintains a stem-cell population through self-renewal; another route produces cells that begin differentiation.
- Place embryonic stem cells on a broad animal-cell branch and adult tissue stem cells on a narrower tissue-related branch, while noting that exact potential depends on type and conditions.
- Place meristem cells on plant branches leading to roots, shoots and specialised plant tissues; do not merge animal potency labels directly with plant development.

Which comparison is best supported by GCSE-level evidence?

- A. Embryonic stem cells can form many animal cell types, while many adult stem cells form a more limited tissue-related range.
- B. Adult stem cells can never divide after birth.
- C. Meristems are specialised animal organs that make blood cells.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Read a source-evidence table using the same headings for every group: source, self-renewal evidence, differentiated outcomes and known limitation.
- Calculate a simple differentiation percentage as cells with the stated marker divided by total observed cells multiplied by 100, keeping it tied to the tested conditions.
- Evaluate a claim only within the evidence: a higher percentage in one study may support greater success under those conditions, but it does not prove a guaranteed treatment or unlimited potential.

Find and repair the misconception

Evidence glitch: because one dish formed nerve-like cells, these cells are proven to cure every nervous-system condition.

A. Keep it: a laboratory cell outcome proves a universal cure.

B. Repair it: the dish supports differentiation under tested conditions, not safety, effectiveness or a guaranteed treatment.

C. Repair it: laboratory evidence can never tell scientists anything useful.

Stem-cell identity and evidence console

Route each evidence card to embryonic stem cells, adult tissue stem cells or plant meristems, then justify one comparison.

- Read the source and outcome evidence on one card.
- Commit it to the matching stem-cell group.
- Repair claims that confuse organism, source or differentiation range.
- Freeze one similarity, one difference and one evidence limit.

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- EMBRYONIC STEM CELLS
- ADULT TISSUE STEM CELLS
- PLANT MERISTEMS

Compare two groups using one shared property, one difference in potential or role and one limit of the evidence.

Your independent mission

Build an evidence-led comparison of embryonic stem cells, adult tissue stem cells and plant meristems, including one percentage calculation and one bounded application claim.

Record: A three-way comparison, correct use of self-renewal/differentiation, one shown percentage calculation and a claim explicitly limited to the supplied evidence.

Choose your support and challenge

- Supported: Sort six cards, complete one similarity-and-difference frame and calculate a percentage from a worked fraction.
- Standard: Compare all three groups, calculate one outcome percentage and explain one biological role using evidence.
- Stretch: Evaluate a claimed application by combining source, potential, quantitative outcome and limitations without moving into unsupported cure language.

Show what you know

Compare embryonic and adult tissue stem cells using differentiation potential and one tissue role.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.